AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science

Faculty of Science and Technology

CSC 01409: PROGRAMMING IN PYTHON

Spring: 2025-2026

Section: [C]

Group No: 6

Project Report On

Project Name [Cash Flow]

Supervised By

MD. Tanzeem Rahat

Submitted By:

Name	ID
1. ALIF HASAN KHAN	23-50186-1
2. MD YEAHYEA JAM	23-50187-1
3. SADMAN SHAKIB	23-50200-1

1. Introduction

Cash Flow CLI is a command-line based financial management system designed to help users track their income and expenses efficiently. It provides a simple and user-friendly interface for managing personal financial records directly from the terminal.

In today's fast-paced world, managing finances is essential. This application offers a lightweight and offline solution that allows users to maintain financial data securely without relying on external platforms.

2. Key Features

- **Add Income:** Users can add income entries with category, amount, and date
- **Add Expense:** Users can record expenses with necessary details
- **View Transactions:** Displays all transactions in a structured table
- **Search Transactions:** Allows searching by category or date
- **Update Transactions:** Modify existing transaction details
- **Delete Transactions:** Remove unwanted entries using transaction ID
- **Summary Report:** Shows total income, total expenses, and net balance
- **Input Validation:** Ensures correct data entry

3. System Architecture

The system follows a modular architecture for better maintainability and scalability.

Module	Responsibility
main.py	The application entry point and main execution loop.
CLI_UI.py/ Features.py	Handles user interaction and terminal formatting.
service.py	Contains core business logic and operation orchestration.
Finance_Service.py	Manages key generation and AES encryption/decryption.
storage.py	Handles JSON persistence and file I/O operations.
utils.py	Helper functions for validation and strength checks.

4. Technologies Used

- **Python 3.x** – Core programming language
- **Colorama** – For colored terminal output
- **Tabulate** – For displaying data in table format
- **JSON** – For data storage

5. How to Run the Project

To get the project running on a local machine, follow these steps in a terminal or command prompt:

1. Clone the repository to your local directory.
   - https://github.com/Shadman-Shakib-Nihan/Python_Project.git

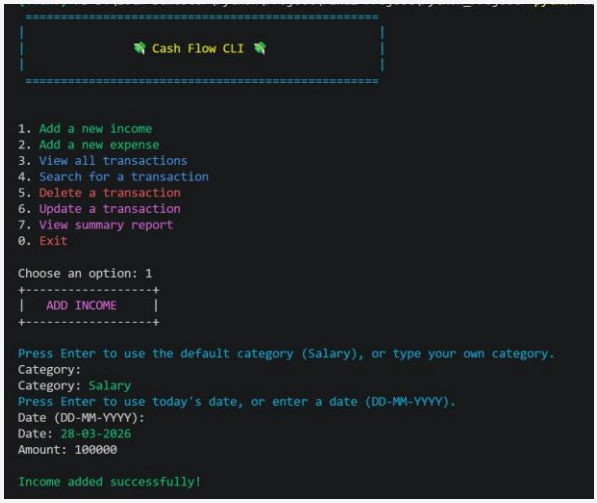
2. Set up a virtual environment (Optional but Recommended):
   - **Windows:**
     - python -m venv .venv
     - .venv\Scripts\activate
   - **macOS/Linux::**
     - python3 -m venv .venv
     - source .venv/bin/activate
3. Install dependencies:
   - pip install -r requirements.txt
4. Run the application:
   - python main.py

6. Conclusion

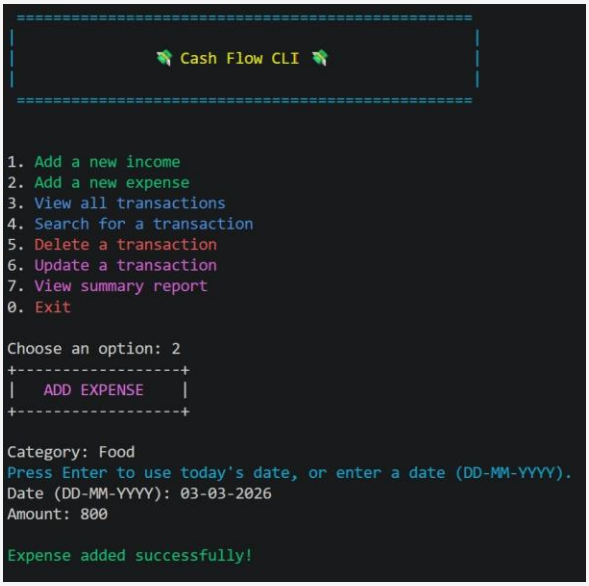
The Cash Flow CLI project successfully demonstrates the use of Python in developing a real-world application. It highlights key programming concepts such as Object-Oriented Programming, file handling, modular design, and validation techniques. This project provides a solid foundation for building more advanced financial systems in the future.

7. Screenshots

7.1 Add a new income



7.2 Add a new expense



7.3 View all transactions

```
Choose an option: 3
```

ID	Category	Amount	Date
2	Salary	60000	2-3-2026
3	Sell	4000	12-03-2026
4	water bill	400	10-03-2026
5	as123	-1000	2026-1-2
6	loan	39930	2-3-2026
7	sdsd	2000	02-03-2026
8	Salary	100000	28-03-2026
9	wifi bill	1200	28-03-2026
10	Salary	10000	28-03-2026

7. 4 Search for a transaction

```
Choose an option: 4

Enter Category/Date(DD-MM-YYYY): Salary
```

ID	Category	Amount	Date
2	Salary	60000	2-3-2026
8	Salary	100000	28-03-2026
10	Salary	10000	28-03-2026
11	Salary	10000	28-03-2026
12	Salary	10000	28-03-2026
14	Salary	5000	28-03-2026
16	Salary	4000	28-03-2026
17	Salary	100000	28-03-2026

7.5 Delete a transaction

```
Choose an option: 5

Enter a category/date (DD-MM-YYYY) to find the transaction you want to delete: Food
```

ID	Category	Amount	Date
13	food	1000	28-03-2026
18	Food	800	03-03-2026

```
Enter Transaction ID to delete: 13
Transaction 13 deleted.
```

7. 6. Update a transaction

```
Choose an option: 6

1.Income
2.Expense
Press 1/2 or Enter type to find the transaction you want to update: 1
```

ID	Category	Amount	Date
2	Salary	60000	2-3-2026
3	Sell	4000	12-03-2026
5	as123	-1000	2026-1-2
6	loan	39930	2-3-2026
8	Salary	100000	28-03-2026
10	Salary	10000	28-03-2026
11	Salary	10000	28-03-2026
12	Salary	10000	28-03-2026
14	Salary	5000	28-03-2026
16	Salary	4000	28-03-2026
17	Salary	100000	28-03-2026

```
Enter Transaction ID to update: 5
New Amount (Current: -1000.0, press enter to keep current): 1000
New Category (Current: as123, press enter to keep current): gift
New Date (DD-MM-YYYY, Current: 2026-1-2, press enter to keep current): 24-03-2026
Transaction 5 updated successfully!
```

7. 7 View summary report

```
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|                                                  |
|              💸 Cash Flow CLI 💸                 |
|                                                  |
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1. Add a new income
2. Add a new expense
3. View all transactions
4. Search for a transaction
5. Delete a transaction
6. Update a transaction
7. View summary report
0. Exit

Choose an option: 7

--- Summary Report ---
Total Income: $343930.00
Total Expenses: -$4420.00
Net Balance: $339510.00
```